Status	Finished
Started	Tuesday, 27 October 2026, 4:58 PM
Completed	Tuesday, 27 October 2026, 5:02 PM
Duration	3 mins 35 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Comparing each element with the others
c. Dividing the input into smaller subproblems
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Insertion sort
c. Heap sort
d. Bubble sort

The correct answer is:
Insertion sort

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are combined to form a single sorted list
c. They are compared element by element
d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It requires external libraries
c. It sorts data out of the box efficiently
d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting decreases the efficiency of selection operations
b. Sorting helps in finding duplicates quickly
c. Sorting makes it harder to search for items
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **6**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. sort()
- c. arrange()
- d. order()

The correct answer is: sorted()

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Removing duplicates from a list
- c. Finding a specific element in a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Merge Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Sorting data in a non-sequential manner
b. Being the most efficient sorting algorithm
c. Bubbling up the largest element to its correct position with each pass
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It divides the list into two halves, sorts each half, and then merges them
c. It builds a sorted array one element at a time
d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **11**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the largest element to its correct position
b. Dividing the list into halves
c. Sorting the entire list in one pass
d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **12**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging

b. Complexity

c. Sorting

d. Searching

The correct answer is:
Complexity

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort

b. Quick Sort

c. Bubble Sort

d. Merge Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Sorting

b. Rearranging

c. Searching

d. Merging

The correct answer is: Sorting

Question **15**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It allows for faster searching
b. It increases the number of comparisons needed
c. It has no effect on the search operation
d. It makes the search operation slower

The correct answer is: It allows for faster searching